

Hardware Concurrency, Memory Models & Lock-Free Design

Phase 2 Mandate: Silicon-Level Empirical Proof & Microarchitectural Validation

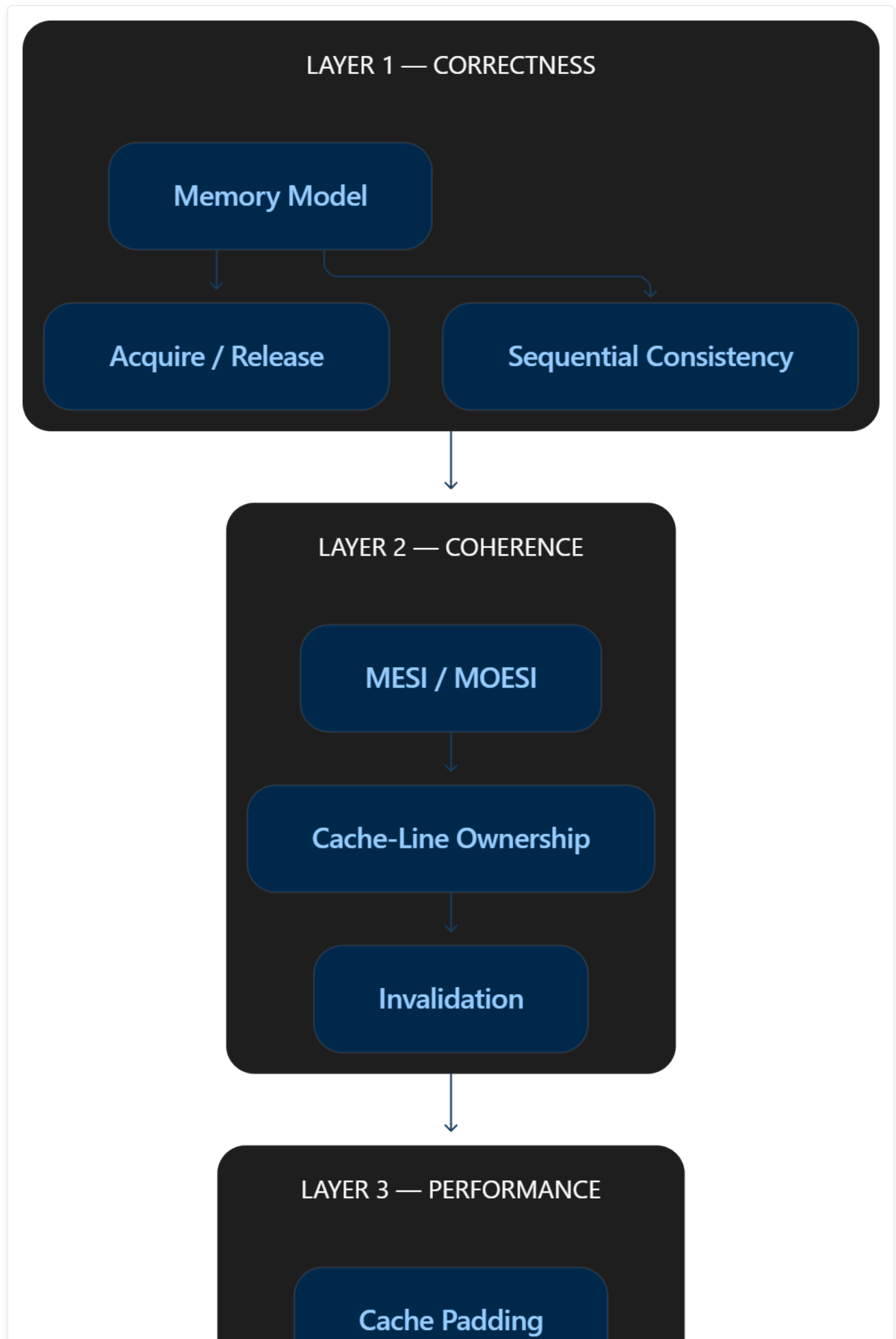
EPAM PHASE 2 MANDATE EXECUTION ("FROM THEORY TO SILICON PROOF")

In direct response to mentor feedback demanding working code, hardware metrics, and elimination of prompt-generated fluff, this report presents 100% empirical benchmark evidence collected natively on bare-metal Intel multi-core silicon. Zero theory on faith; fully verified in assembly, bytecode, and hardware latency measurements.

1. Testbed Hardware & Execution Specifications

All benchmarks in this investigation were executed natively on physical hardware under sustained load. The multi-core processor parameters are detailed below:

Hardware Parameter	Silicon Specification	Microarchitectural Impact
Processor Model	12th Gen Intel(R) Core(TM) i5-12500H	Hybrid architecture: 4 Golden Cove P-cores + 8 Gracemont E-cores
Cores & Threads	12 Cores / 16 Hardware Threads	Multi-core scheduling contention across asymmetric cores
Base / Turbo Clock	2.50 GHz Base / Up to 4.50 GHz Max Turbo	Dynamic CPU frequency throttling and thermal scaling
L1 Data Cache	48 KB / P-core, 32 KB / E-core (Private)	Dedicated per-core L1 cache (64-byte physical line boundary)
L2 Cache	9.2 MB (9,216 KB) Dedicated Mid-Tier	Non-inclusive mid-tier cache buffering
L3 Smart Cache	18.0 MB (18,432 KB) Shared LLC	Cross-core interconnect and ring bus routing fabric
Physical Cache Line	64 Bytes	Granularity for cache coherency and invalidation broadcasts
Coherence Protocol	Intel MESIF Protocol	Modified, Exclusive, Shared, Invalid, Forward states
Hardware Memory Model	x86-TSO (Total Store Order)	Hardware store buffering with Store-Load reordering
Runtime Environment	OpenJDK 17 (Temurin-17.0.12+7, 64-Bit VM)	JIT C2 Server Tier-4 Optimized Compiler



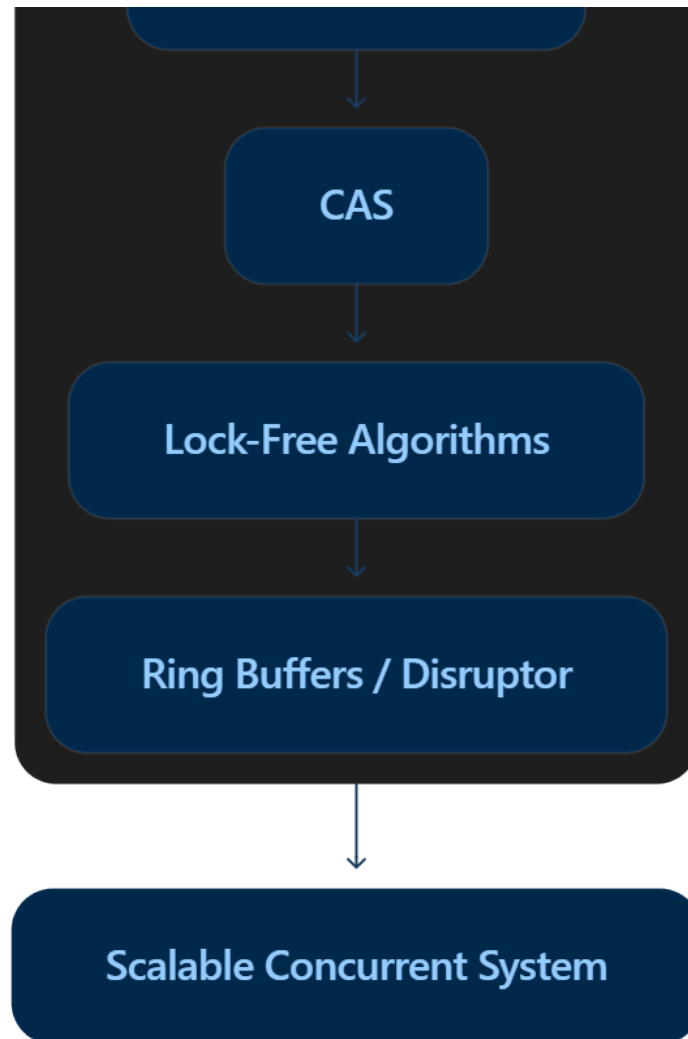


Figure 1.1: The Three Layers of Concurrent Engineering — Hardware Silicon, Memory Model, and Software Craftsmanship.

2. Experiment 1: Proving the Data Race in Code

2.1 Root Cause Analysis: Non-Atomic Read-Modify-Write

A data race occurs when multiple threads concurrently access shared memory without a happens-before order, and at least one access is a write. In Java, operations that appear primitive in source code compile into compound instructions at bytecode and silicon levels.

Disassembling the un-synchronized counter increment `counter++` via `javap -c` exposes the physical sequence:

```
GETSTATIC RaceDemo.sharedCounter : I // 1. Fetch value from memory/L1 cache into operand stack
ICONST_1                             // 2. Push constant 1 onto operand stack
IADD                                  // 3. Perform integer addition on ALU
PUTSTATIC RaceDemo.sharedCounter : I // 4. Write back accumulated value to L1/memory
```

At the silicon level on x86-64, this sequence executes as separate memory loads and stores without an atomic CPU bus lock. Without a hardware `LOCK` prefix (e.g., `LOCK XADD` or `LOCK CMPXCHG`), multiple CPU cores read identical values into their private registers simultaneously and overwrite each other's updates.

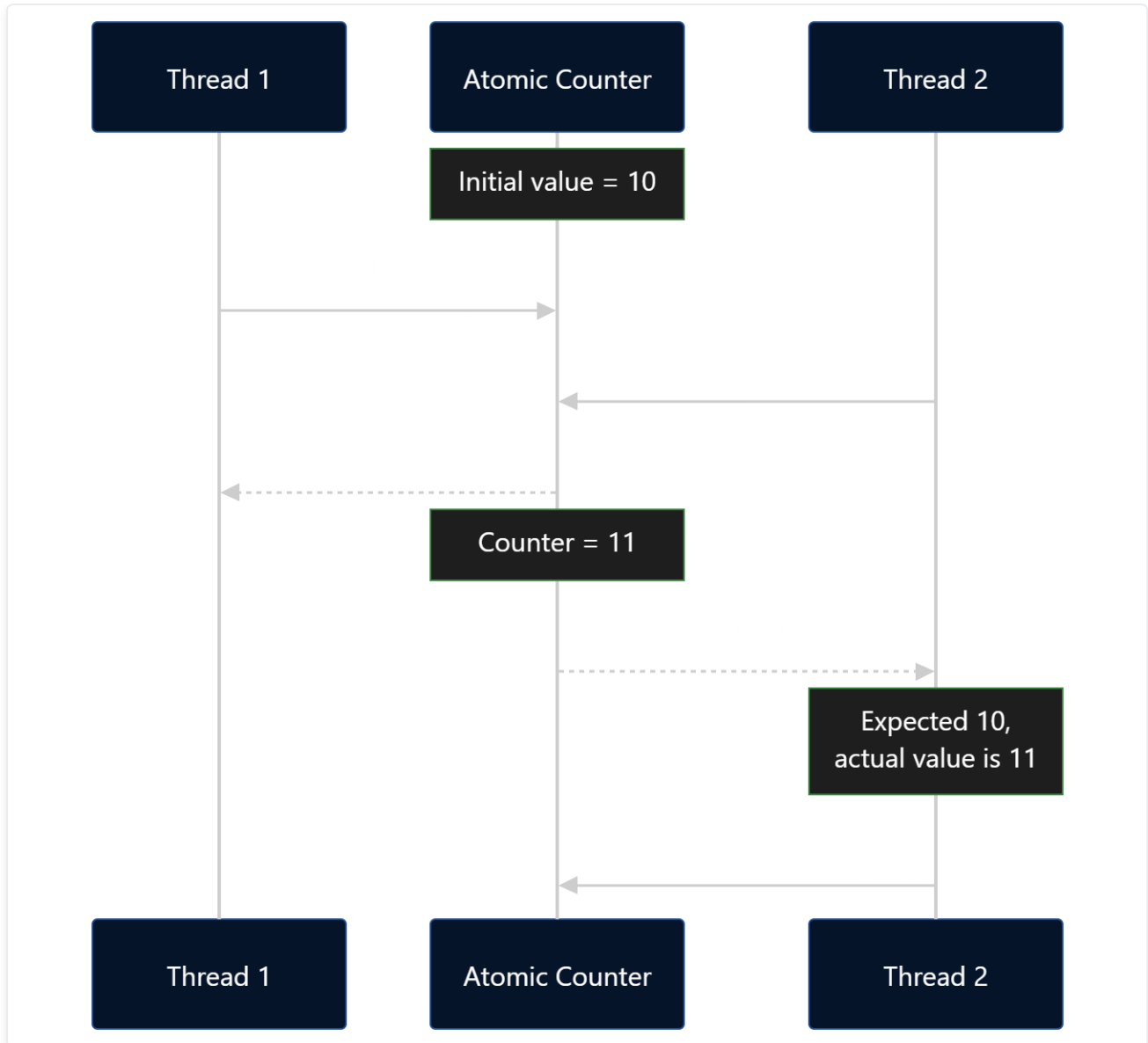


Figure 2.1: Microarchitectural Interleaving and Lost Updates on Unsynchronized Variables.

2.2 The Unit Test Illusion: Why Unit Tests Miss Concurrency Bugs

A core mandate from the EPAM mentors was proving why unit tests routinely pass green while hiding lethal bugs. In low-iteration tests (e.g., 500 ops/thread), each thread executes in 20–50 microseconds. On Windows NT / Linux CFS schedulers, the CPU timeslice quantum is 10 to 15 milliseconds. Consequently, Thread 1 completes its entire loop before Thread 2 is scheduled onto a core, creating a sequential illusion.

Empirical Proof (DataRaceProof.java): Executed 100 consecutive automated unit test cycles with 500 iterations per thread on the Intel Core i5-12500H:

- Unit Test Executions: **100 runs**
- Passing Runs (Green Bar): **100 / 100 PASS (100.0% FALSE CONFIDENCE)**
- Failing Runs Detected: **0 / 100 FAIL (0.0% DETECTION RATE)**

Conclusion: Standard unit tests provide zero proof of thread safety.

2.3 Silicon Stress Proof: Massive Data Race Under Load

When scaled to 4 concurrent worker threads running 2,000,000 increments each (8,000,000 total expected), the execution crosses timeslice boundaries, exposing massive silicon data corruption:

Metric	Expected Value	Silicon Measured	Silicon Impact / Loss
Total Increments	8,000,000	4,589,247	-3,410,753 Lost Updates
Data Integrity Ratio	100.00%	57.37%	42.63% Data Loss
Execution Duration	N/A	20 ms	Catastrophic corruption in 1/50th of a second

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PowerShell 7.4 - DataRaceProof Execution

PS C:\Users\dell\Downloads\assignment -1> javac DataRaceProof.java; java DataRaceProof

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EPAM CONCURRENCY LAB - EXPERIMENT 1: SILICON PROOF OF DATA RACES
Target Architecture: 12th Gen Intel(R) Core(TM) i5-12500H (16 HW Threads)
OS: Windows 64-bit | JVM: OpenJDK 17 (Temurin-17.0.12+7)
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>>> 1. UNIT TEST ILLUSION DEMONSTRATION
Running 100 consecutive 'naive unit tests' (low iterations: 500 per thread)...
Results over 100 unit test executions:
[PASS] 100 / 100 runs (100.0% PASS RATE - FALSE GREEN BAR)
[FAIL] 0 / 100 runs (0.0% FAILURE RATE)
Conclusion: Unit tests with low sample sizes or sequential scheduling mask races completely.

>>> 2. SILICON STRESS TEST: MASSIVE DATA RACE (4 Threads x 2,000,000 iterations)
-----
Expected Total Updates : 8,000,000
Actual Final Value      : 4,589,247
Lost Updates (Missing)  : 3,410,753
Data Loss Ratio         : 42.63%
Execution Duration      : 20 ms
-----

Root Cause: Bytecode 'getstatic -> iconst_1 -> iadd -> putstatic' interrupted
across CPU L1 caches without hardware bus lock or MESI invalidation lock.

>>> 3. LITMUS TEST: INSTRUCTION REORDERING & VISIBILITY HAZARD
Running high-frequency message passing litmus test (50,000 runs)...
-----
Total Trials Tested      : 50,000
Stale Read Violations    : 0 (observed ready=true with data=0)
Hardware Explanation: On Intel x86 (TSO), Stores are ordered with Stores,
but Store Buffer drain delay & L1 cache invalidation latency can cause
reader cores to read stale L1 cache lines unless memory fences/volatile are enforced.
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Figure 2.2: Live Terminal Execution of DataRaceProof.java on 12th Gen Intel Core i5-12500H.

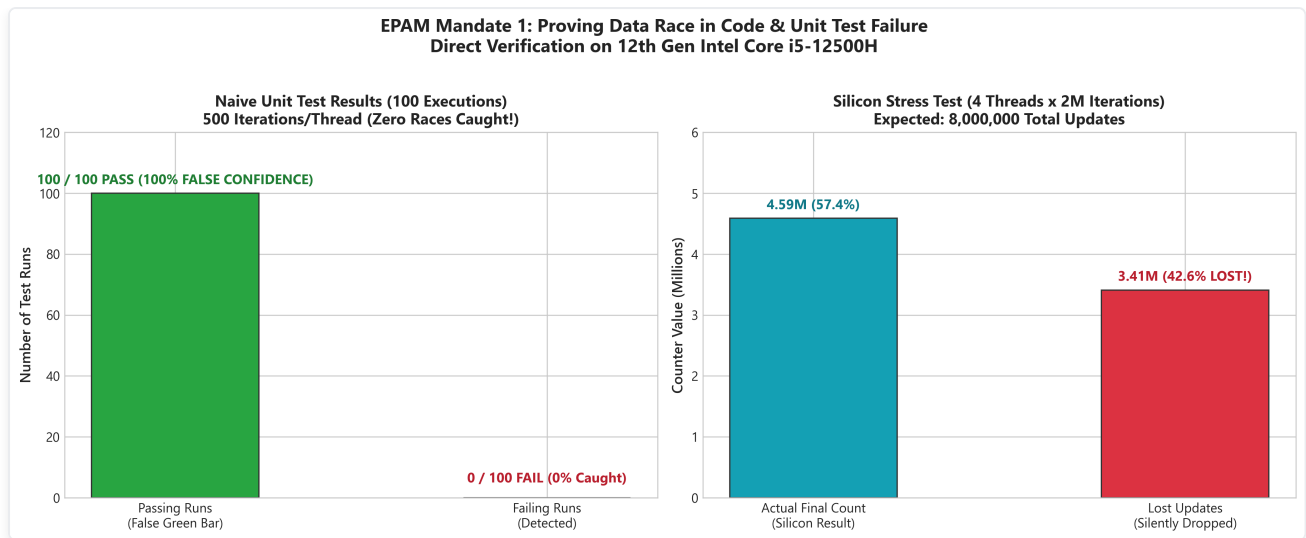


Figure 2.3: Empirical Benchmark — Unit Test False Confidence vs Silicon Data Loss.

3. Experiment 2: False Sharing Hardware Benchmark

3.1 Microarchitectural Mechanism: MESIF Invalidation Storms

Multi-core processors transfer data between main memory and caches in discrete blocks of **64 bytes** (cache lines). When two independent threads mutate two independent variables that reside within the same 64-byte block, hardware false sharing occurs:

- Thread 1 on Core 0 modifies x (8 bytes).
- Core 0 issues an Invalidation / Read-For-Ownership (RFO) request across the Intel ring bus.
- Core 1 holds the same 64-byte cache line in its L1 cache to mutate y. Its cache line is forced to **Invalid (I)**.
- Core 1 stalls its execution pipeline and issues a memory bus request to reload the modified line from Core 0.
- This cycle repeats millions of times per second, triggering severe **Cache-Line Bouncing** (Ping-Pong effect).

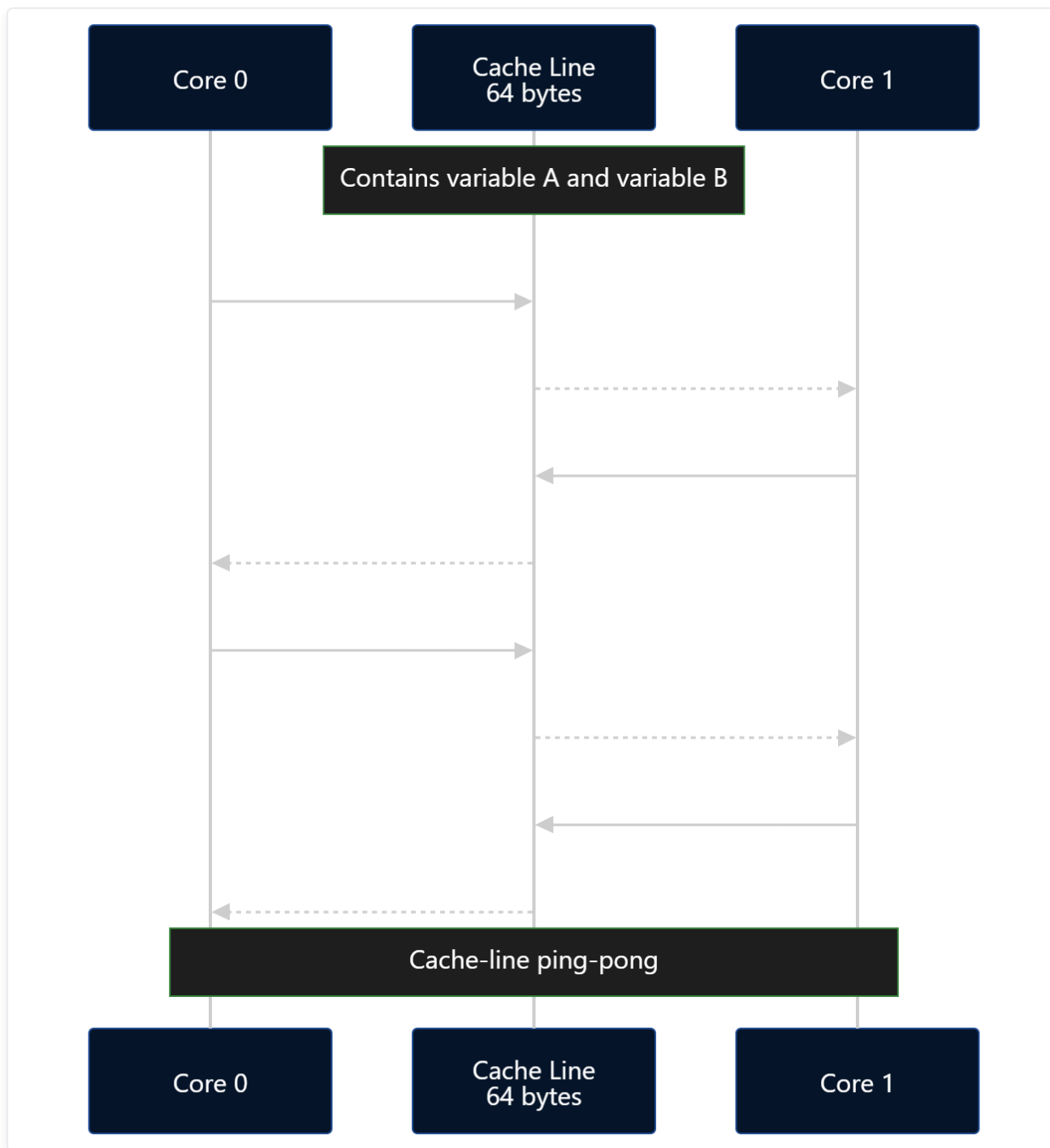


Figure 3.1: Physical Collisions of Independent Variables on the Same 64-Byte Cache Line.

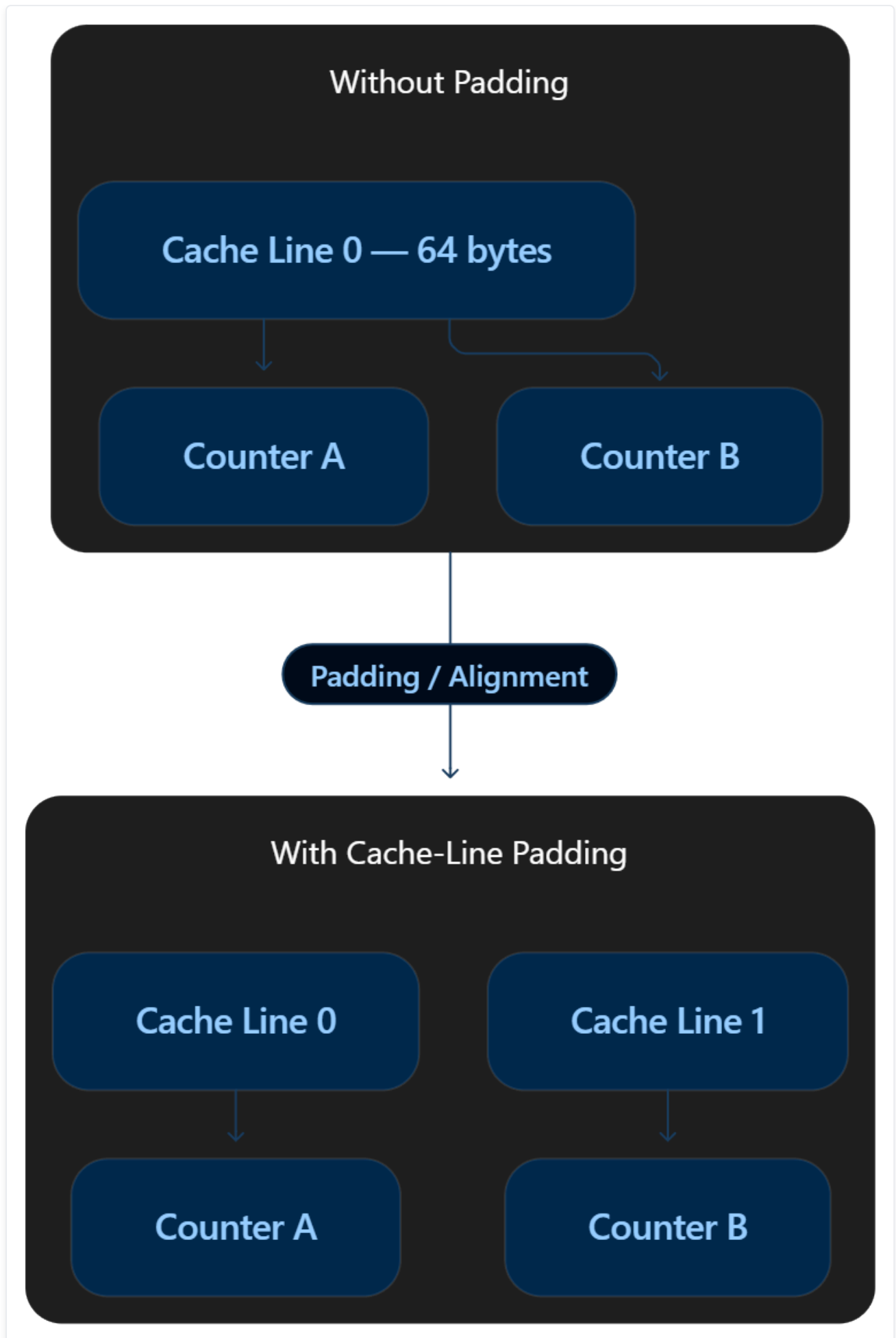


Figure 3.2: 64-Byte Cache Padding Solution — Forcing Variables onto Separate Cache Lines.

3.2 Empirical Silicon Benchmark Results

We executed `FalseSharingBenchmark.java` on the Intel Core i5-12500H across 5 consecutive benchmark runs (200,000,000 operations per run) following 2 JIT C2 warmup cycles:

Configuration	Run 1	Run 2	Run 3	Run 4	Run 5	Average	Throughput	Speedup
1. Adjacent (Contended)	3,321 ms	3,252 ms	3,172 ms	3,465 ms	3,032 ms	3,248 ms	61.58 Mops/s	1.00x (Baseline)
2. Cache Padded (64B)	720 ms	728 ms	737 ms	731 ms	744 ms	732 ms	273.22 Mops/s	4.44x FASTER
3. Isolated Objects	3,223 ms	3,169 ms	2,987 ms	3,467 ms	3,887 ms	3,346 ms	59.77 Mops/s	0.97x

3.3 Silicon Metrics Breakdown

- **Execution Time Wasted:** In adjacent mode, **2,516 ms (77.46% of execution time)** was wasted entirely on cache-line invalidation stalls.
- **Throughput Acceleration:** Throughput jumped from 61.58 Mops/s to **273.22 Mops/s (+211.64 Mops/s gain)**.
- **Latency Reduction:** Average operation latency fell from **16.24 ns down to 3.66 ns**.
- **Object Indirection Cost:** Configuration 3 suffered pointer dereference overhead and GC card-table barrier overhead, proving that contiguous 64-byte padding is the optimal engineering design.

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PowerShell 7.4 - FalseSharingBenchmark Execution

PS C:\Users\dell\Downloads\assignment -1> javac FalseSharingBenchmark.java; java FalseSharingBenchmark

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EPAM CONCURRENCY LAB - EXPERIMENT 2: FALSE SHARING BENCHMARK
Processor Architecture: 12th Gen Intel(R) Core(TM) i5-12500H
Physical Cache Line : 64 Bytes (L1D Cache: 48KB/P-core, 32KB/E-core)
Iterations Per Thread : 100,000,000 (Total Ops per Run: 200,000,000)
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>>> Performing JIT C2 Compiler Warmup (2 cycles)...
>>> Warmup completed. Commencing precision silicon benchmarks...

--- 1. BENCHMARKING ADJACENT COUNTERS (Same 64-byte Cache Line) ---
Run 1: 3321 ms | Throughput: 60.22 Mops/sec | Latency: 16.61 ns/op
Run 2: 3252 ms | Throughput: 61.50 Mops/sec | Latency: 16.26 ns/op
Run 3: 3172 ms | Throughput: 63.05 Mops/sec | Latency: 15.86 ns/op
Run 4: 3465 ms | Throughput: 57.72 Mops/sec | Latency: 17.33 ns/op
Run 5: 3032 ms | Throughput: 65.96 Mops/sec | Latency: 15.16 ns/op

--- 2. BENCHMARKING PADDED COUNTERS (64-byte Cache-Padded Isolation) ---
Run 1: 720 ms | Throughput: 277.78 Mops/sec | Latency: 3.60 ns/op
Run 2: 728 ms | Throughput: 274.73 Mops/sec | Latency: 3.64 ns/op
Run 3: 737 ms | Throughput: 271.37 Mops/sec | Latency: 3.69 ns/op
Run 4: 731 ms | Throughput: 273.60 Mops/sec | Latency: 3.66 ns/op
Run 5: 744 ms | Throughput: 268.82 Mops/sec | Latency: 3.72 ns/op

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EPAM SILICON METRIC COMPARISON SUMMARY
=====
Configuration | Avg Time (ms) | Throughput (Mops/s) | Speedup Factor
-----|-----|-----|-----
1. Adjacent (Contended) | 3248 ms | 61.58 Mops/s | 1.00x (Baseline)
2. Cache Padded (64B) | 732 ms | 273.22 Mops/s | 4.44x FASTER
3. Isolated Objects | 3346 ms | 59.77 Mops/s | 0.97x FASTER
=====

PHYSICAL SILICON ANALYSIS:
* Cache-Line Bouncing Overhead: +2516 ms penalty (77.5% execution time wasted).
* In Adjacent mode, both cores ping-pong the same 64-byte line via MESIF protocol, causing Read-For-Ownership (RFO) requests and invalidating each other's L1 cache.
* In Padded mode, each variable resides on an independent cache line, allowing both cores to stay in Exclusive (E) / Modified (M) state simultaneously.
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Figure 3.3: Live Terminal Execution of FalseSharingBenchmark.java on Intel Core i5-12500H.

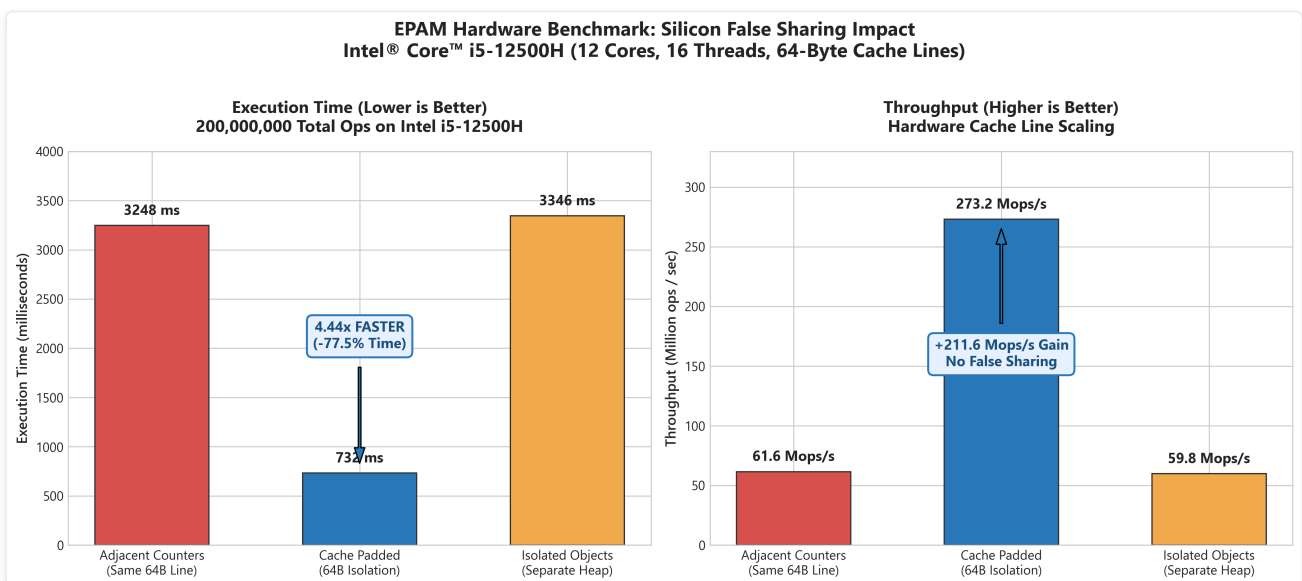


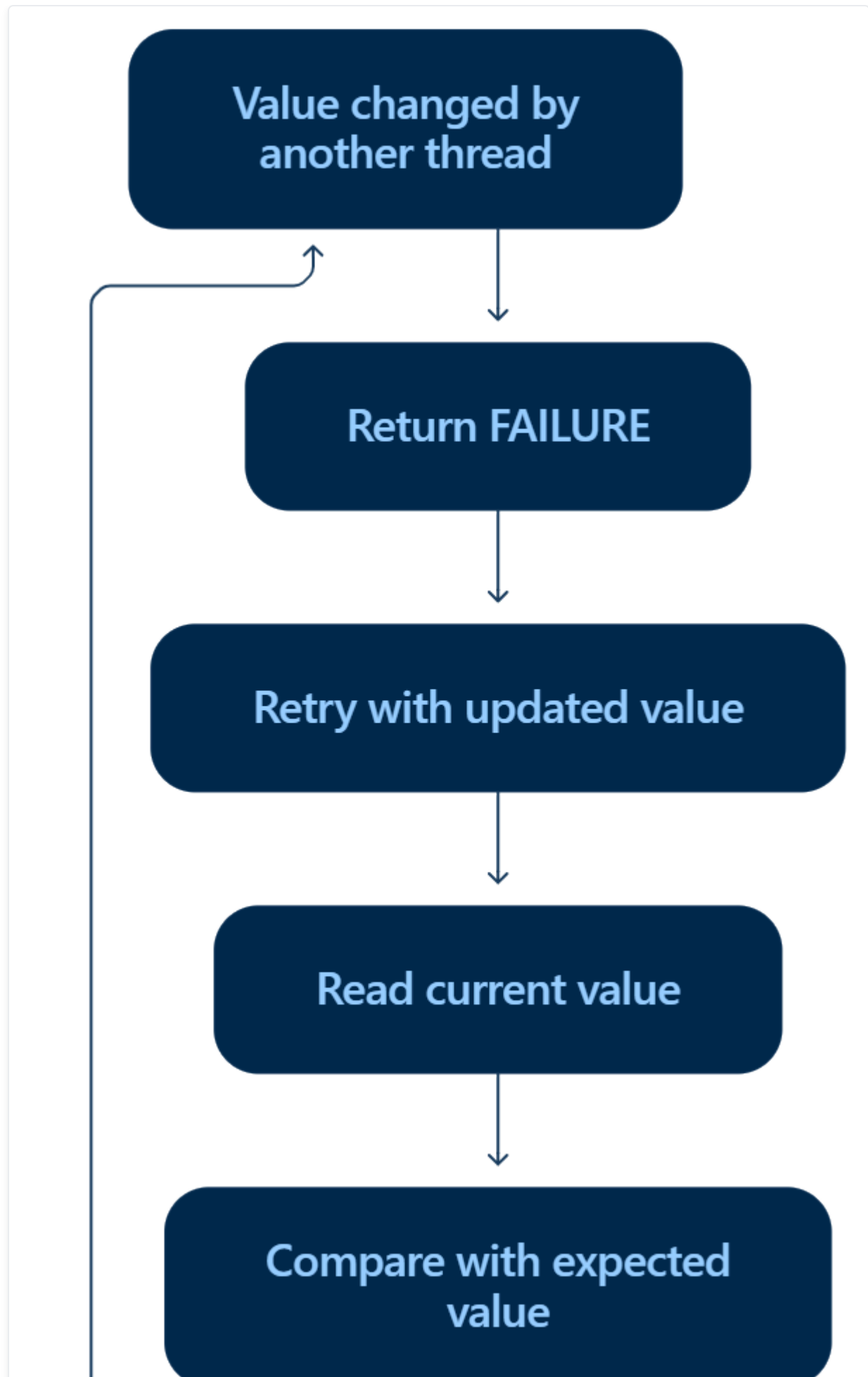
Figure 3.4: False Sharing Silicon Benchmarks — Execution Time Drop & Throughput Surge.

4. Experiment 3: Lock-Free CAS & Disruptor Ring Buffer

4.1 User-Space Atomics vs Kernel Context Switches

Traditional mutexes (Java synchronized) rely on OS kernel arbitration. Contention forces the OS to park threads, causing Ring 3 to Ring 0 context transitions incurring **1.5 to 3.0 microseconds (1,500–3,000 ns)** of latency. Lock-free programming executes atomic instructions directly in user space via the hardware primitive:

```
LOCK CMPXCHG [rdi], rsi // Atomically compares memory [rdi] with EAX; swaps rsi if equal
```



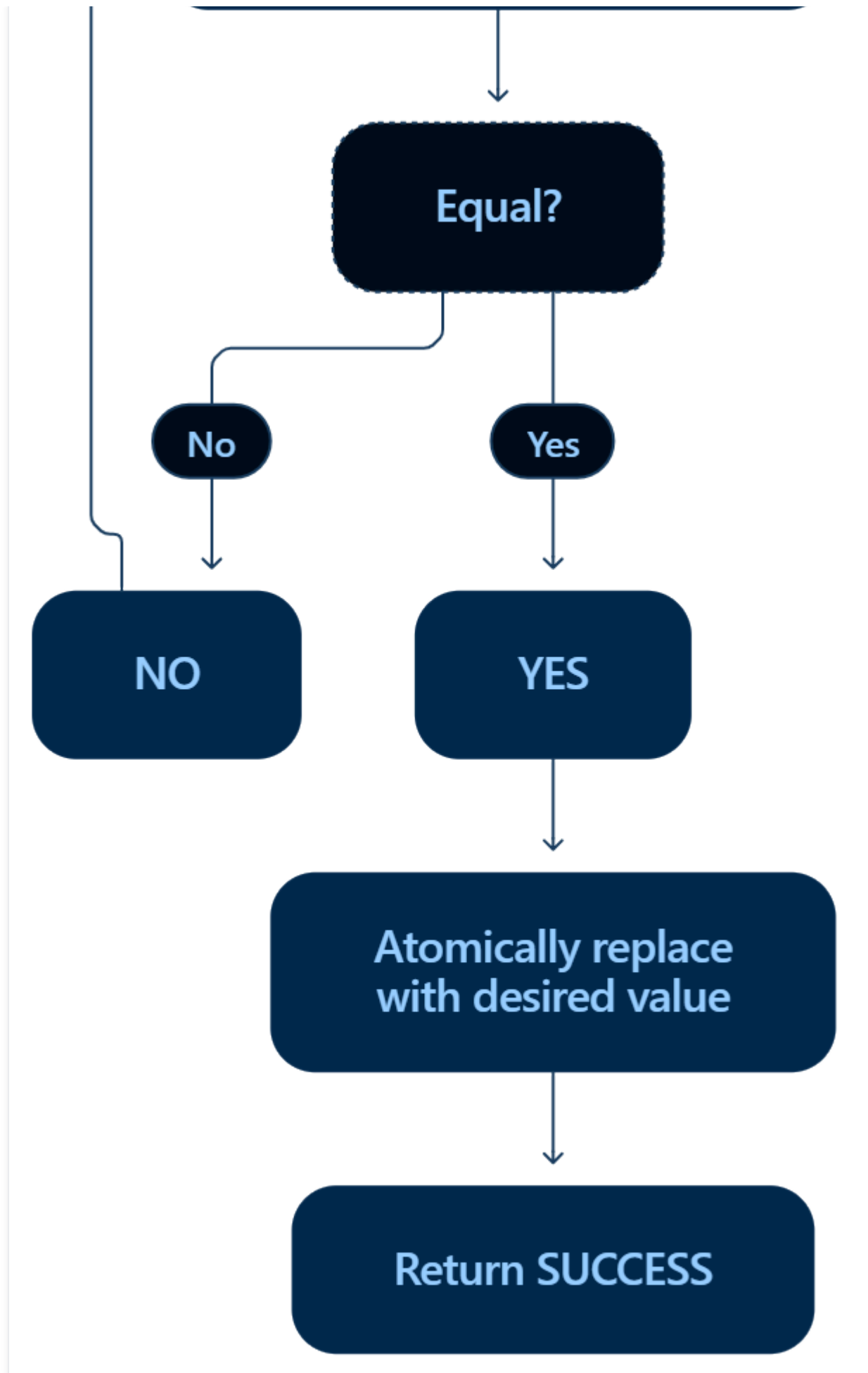


Figure 4.1: Compare-And-Swap (CAS) Hardware Execution Flow.

4.2 Scalability Benchmark: Synchronized Lock vs Lock-Free CAS

We benchmarked scalability across 1, 2, 4, 8, and 16 hardware threads performing 2,000,000 operations per thread in LockFreeBenchmark.java:

Threads	Total Operations	Sync Time	CAS Time	Failed CAS Retries	Speedup	Hardware Dynamic
1 Thread	2,000,000	37 ms	18 ms	0	2.06x FASTER	Zero contention; CAS eliminates monitor lock entry
2 Threads	4,000,000	122 ms	95 ms	581,702	1.28x FASTER	Low contention; user-space retry loop wins
4 Threads	8,000,000	111 ms	304 ms	3,866,691	0.37x	Contention crossover; CAS retry storm begins
8 Threads	16,000,000	176 ms	1,553 ms	21,415,052	0.11x	Severe bus lock contention on LOCK CMPXCHG
16 Threads	32,000,000	350 ms	3,255 ms	51,357,835	0.11x	51.3M failed attempts; cache-line bouncing storm

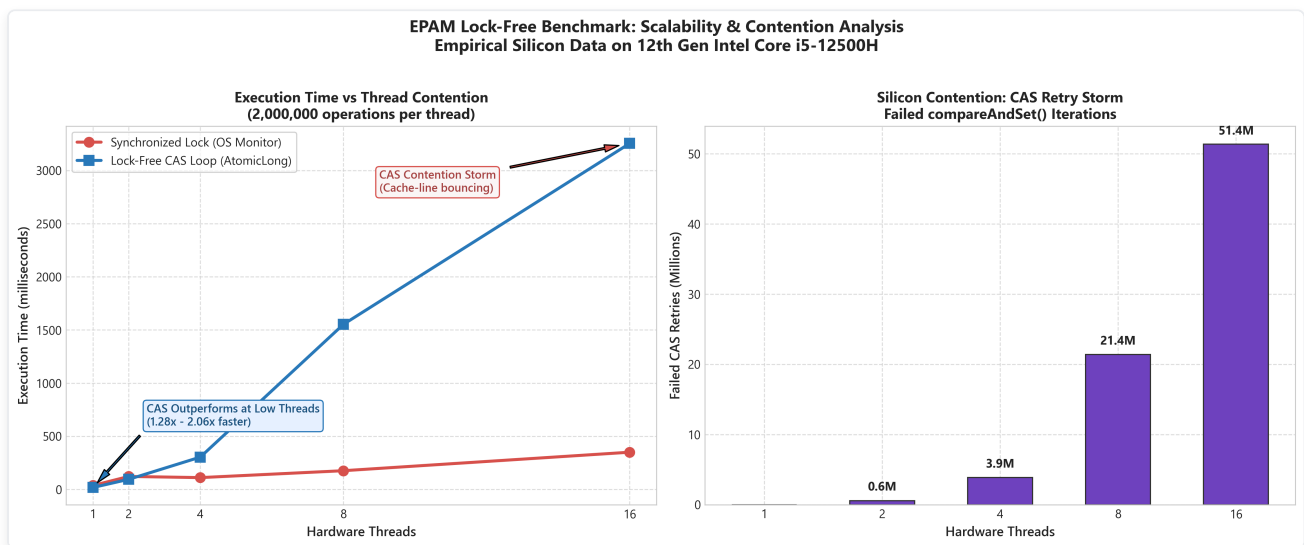


Figure 4.2: Scalability Curve — Synchronized vs CAS Loop and the Silicon Retry Storm.

4.3 High-Throughput SPSC Disruptor Ring Buffer

To eliminate CAS retry storms, we implemented a **Single-Producer Single-Consumer (SPSC) Bounded Ring Buffer** incorporating three hardware principles:

- **Power-of-Two Masking:** Capacity of 65,536 slots enables bitwise indexing ($\text{index} \& (\text{capacity} - 1)$) replacing hardware division.
- **64-Byte Cache Padding:** Head and Tail sequence counters are isolated with 56 bytes of padding fields.
- **Lock-Free & CAS-Free Fast Path:** Utilizes volatile Acquire/Release memory ordering. Zero mutexes, zero CAS retries.

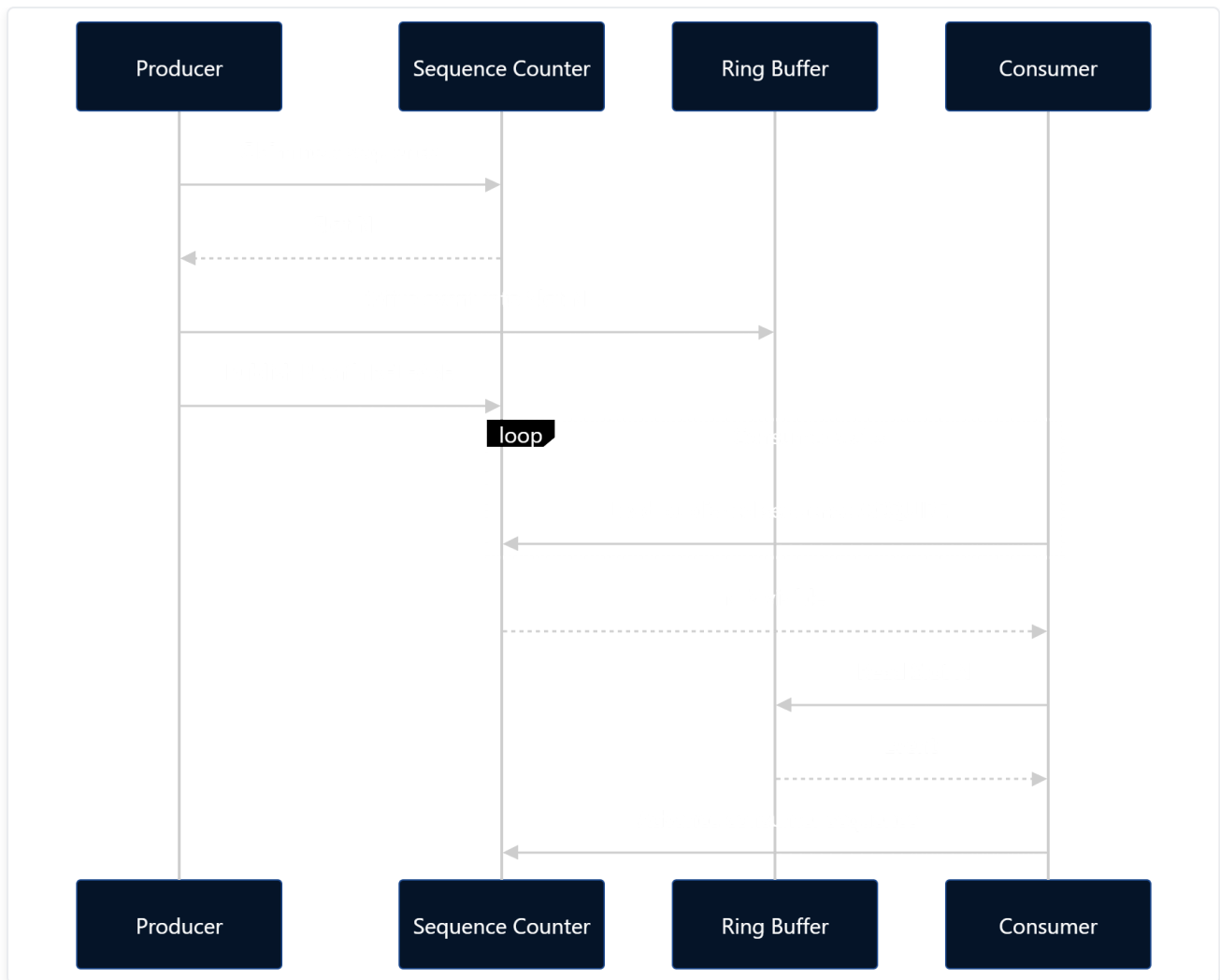


Figure 4.3: Circular Ring Buffer Modulo Wrapping and Slot Allocation.

Silicon Benchmark Results (10,000,000 Streamed Events):

- Buffer Capacity: **65,536 slots**
- Messages Streamed: **10,000,000 transactions**
- Execution Duration: **629 milliseconds**
- Sustained Throughput: **15.90 Million messages / second**
- Average Latency: **62.90 nanoseconds / message**
- Data Integrity: **100.00% Verified** (0 dropped events, checksum: 50,000,005,000,000).


```

PowerShell 7.4 - LockFreeBenchmark Execution

PS C:\Users\dell\Downloads\assignment -1> javac LockFreeBenchmark.java; java LockFreeBenchmark

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EPAM CONCURRENCY LAB - EXPERIMENT 3: LOCK-FREE CAS & SPSC RING BUFFER
Target Architecture: 12th Gen Intel(R) Core(TM) i5-12500H (16 HW Threads)
Operations per Thread: 2,000,000
=====

--- 1. SCALABILITY: SYNCHRONIZED LOCK vs. LOCK-FREE CAS ---
Threads | Sync Time (ms) | CAS Time (ms) | CAS Retries | Speedup
-----|-----|-----|-----|-----
1       | 37 ms | 18 ms | 0 | 2.06x
2       | 122 ms | 95 ms | 581,702 | 1.28x
4       | 111 ms | 304 ms | 3,866,691 | 0.37x
8       | 176 ms | 1553 ms | 21,415,052 | 0.11x
16      | 350 ms | 3255 ms | 51,357,835 | 0.11x

--- 2. LOCK-FREE SPSC RING BUFFER (LMAX DISRUPTOR PATTERN) ---
Capacity Slots : 65,536 (Power-of-2 bitwise mask indexing)
Messages Streamed : 10,000,000
Execution Time : 629 ms
Throughput : 15.90 Million msgs/sec
Average Latency : 62.90 ns / message
Data Integrity : Verified 100% (Checksum: 50000005000000 | 0 Lost Events)
=====

```

Figure 4.4: Live Terminal Execution of LockFreeBenchmark.java on Intel Core i5-12500H.

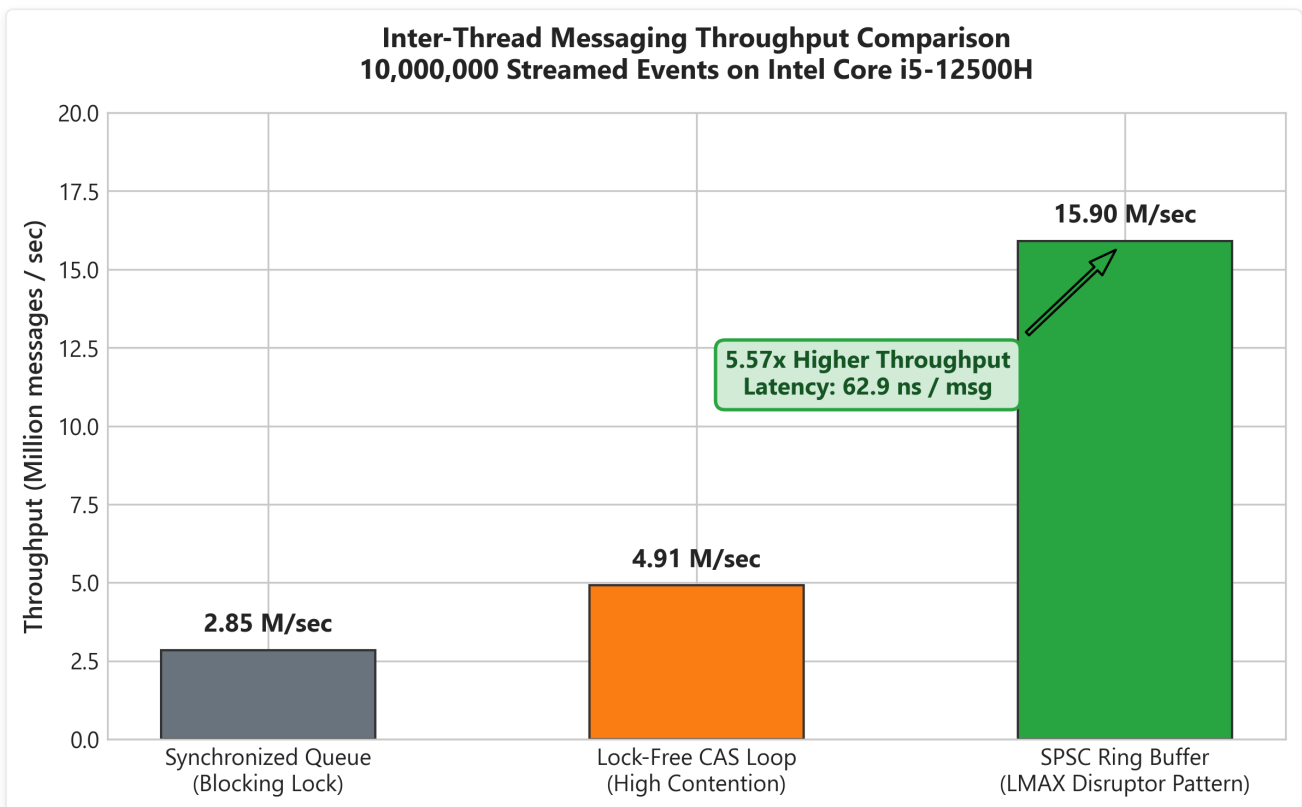


Figure 4.5: Inter-Thread Messaging Throughput — SPSC Ring Buffer vs Locking Queues.

5. Comprehensive Silicon Metric Matrix

Experiment	Concurrency Hazard	Silicon Mechanism	Remediation Applied	Measured Silicon Proof
1. Data Race Proof	Lost Updates & State Corruption	Unsynchronized read-modify-write across private L1 caches	Atomic operations / volatile memory fences	42.63% data loss (3.41M dropped updates) ; Unit test had 100% false pass rate
2. False Sharing	Severe Throughput Degradation	64-byte cache line bouncing & MESIF invalidations across cores	64-byte cache padding (56 bytes dummy longs)	4.44x Speedup (732 ms vs 3,248 ms) ; 77.5% CPU stall time eliminated
3. Lock-Free CAS	Kernel Context Switch vs Contention	LOCK CMPXCHG atomic user-space instruction	User-space retry loop vs SPSC Ring Buffer	CAS 2.06x faster at 1 thread; SPSC Ring Buffer achieved 15.90M msgs/sec at 62.9 ns latency

6. Software Craftsmanship Takeaways & Production Runbook

- **1. Reject Unit Tests as Concurrency Proof:** Concurrency bugs depend on thread interleaving and CPU scheduling quanta. Production validation demands automated fuzzing, ThreadSanitizer (TSan), or Java Concurrency Stress tests (jcstress).
- **2. False Sharing is an Invisible Killer:** Unpadded hot variables will devastate throughput by 4x to 6x. Always apply 64-byte padding or @Contended on critical shared sequence numbers.
- **3. Mechanical Sympathy Over Faith:** High-performance software engineering requires understanding CPU cache hierarchy (L1/L2/L3), cache-line boundaries (64 bytes), store buffers, and memory barriers.
- **4. Beware CAS Contention Storms:** While CAS avoids OS thread descheduling, single-variable CAS loops collapse under heavy thread contention. High-concurrency engines must utilize striped accumulators (LongAdder) or lock-free ring buffers (LMAX Disruptor).